DHVANI Shield v1 – PCB Design Brief

Oct 4, 2026 · Nucleo-144 (STM32U575) shield · SIH 2026 · PS 26058

Goal and rules

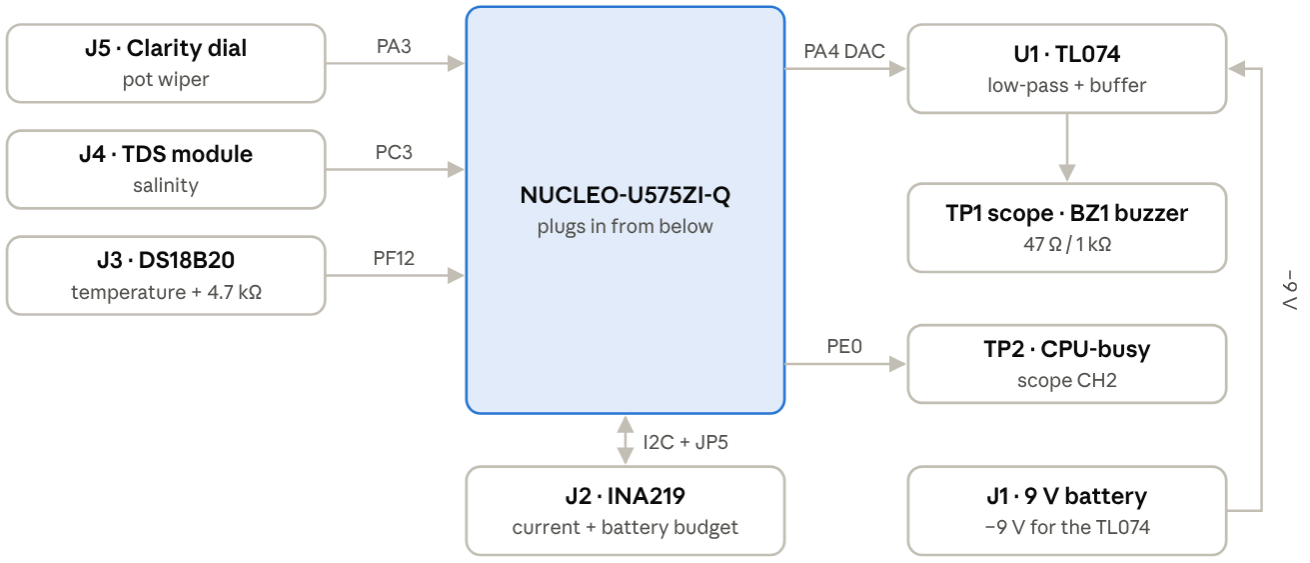
Design one PCB that plugs onto the NUCLEO-U575ZI-Q and replaces the breadboard. It is for the video and roadmap only — it will not be fabricated now.

- **Board:** shield for the NUCLEO-U575ZI-Q (Nucleo-144 headers CN7, CN8, CN9, CN10).
- **Circuit:** exactly what is on the breadboard today — no new parts, no pin changes.
- **Tool:** EasyEDA (free, in the browser; TL074 and connectors are in its library; one-click 3D view).
- **Layers:** 2-layer board.
- **Label everywhere:** "Phase 2: designed, not yet fabricated". Do not claim it works on other Nucleo boards.

Block diagram

Sensors feed the Nucleo on the left; the DAC signal goes out through the filter on the right.

DHVANI shield: what connects to what



5 V and 3.3 V come from the Nucleo headers (CN8); all grounds are common.

shield blocks · sensors in on the left, signal out on the right

Parts list (BOM)

Use through-hole parts if unsure — they look clear in the 3D view.

Ref	Part	Value / note
U1	TL074 quad op-amp	DIP-14 (or SOIC-14)
R1, R2	Resistor	3.3 kΩ — filter
R3	Resistor	47 Ω — scope output
R4	Resistor	1 kΩ — buzzer
R5	Resistor	4.7 kΩ — DS18B20 pull-up
C1	Capacitor	200 pF (or 2 × 100 pF in parallel)
C2	Capacitor	100 pF
C3, C4	Capacitor	100 nF — op-amp supply decoupling
BZ1	Buzzer	Passive, through-hole
J1	2-pin connector	9 V battery
J2	6-pin header	INA219 module (VCC, GND, SDA, SCL, VIN+, VIN−)
J3	3-pin screw terminal	DS18B20 probe (VDD, DATA, GND)
J4	3-pin header	TDS module (+, −, A)
J5	3-pin header	Clarity dial pot (3V3, wiper, GND)
J6	2-pin header	To Nucleo JP5 (current path)
TP1–TP3	Test points	Scope CH1, scope CH2, GND
—	Nucleo-144 headers	CN7, CN8, CN9, CN10 (ST Zio layout)

Connections (netlist)

These match the breadboard exactly. Draw the schematic from this table.

Net	Nucleo pin	Goes to on the shield
DAC_OUT	CN7-9 (PA4)	R1 → node X → R2 → U1 pin 5
Filter caps	—	X → C1 → U1 pin 7; U1 pin 5 → C2 → GND

Net	Nucleo pin	Goes to on the shield
Filter feedback	—	U1 pin 6 ↔ pin 7
Buffer	—	U1 pin 7 → pin 10; pin 9 ↔ pin 8
OUT_SCOPE	—	U1 pin 8 → R3 (47 Ω) → TP1
OUT_BUZZ	—	U1 pin 8 → R4 (1 kΩ) → BZ1 → GND
+5V	CN8-9	U1 pin 4; C3 to GND
−9V	J1 battery −	U1 pin 11; C4 to GND; battery + → GND
Unused amps	—	pin 3 → GND, pin 2 ↔ 1; pin 12 → GND, pin 13 ↔ 14
3V3	CN8-7	J2 VCC, J3 VDD, J4 +, J5 top, R5
GND	CN8-11, CN7-8	All grounds, TP3, ground pour
TEMP_DATA	CN7-20 (PF12)	J3 DATA; R5 to 3V3
TDS	CN9-5 (PC3)	J4 A
CLARITY	CN9-1 (PA3)	J5 wiper
I2C_SDA	CN7-4 (PB9)	J2 SDA
I2C_SCL	CN7-2 (PB8)	J2 SCL
I_SENSE	JP5 pins 1 and 2	J6 → J2 VIN+ / VIN−
CPU_BUSY	CN10-33 (PE0)	TP2

Steps in EasyEDA

About 1–2 days of work.

1. **New project** → name it "DHVANI Shield v1".
2. **Schematic:** place the parts from the BOM (search "TL074", "pin header", "screw terminal", "buzzer"), then wire them using the netlist. Use net labels (DAC_OUT, 3V3, GND…) to keep it tidy.
3. **Check:** run Design → Check DRC/ERC and fix any unconnected pins.
4. **PCB:** convert to PCB. Set the board outline to the Nucleo-144 footprint size and place the CN7–CN10 headers at the Nucleo's header positions.
5. **Place parts:** filter (U1, R1, R2, C1, C2) together near the PA4 pin; C3/C4 right next to U1 pins 4 and 11; connectors J1–J6 along the board edges; TP1–TP3 at the front edge.
6. **Route:** short, direct signal traces; 0.5 mm or wider for power (5 V, 3V3, −9 V).

7. **Ground:** add a solid GND copper pour on the bottom layer.
8. **Silkscreen:** "DHVANI Shield v1 · SIH 2026 · PS 26058" plus labels on every connector (DIAL, TDS, TEMP, INA219, 9V, SCOPE).
9. **3D view:** check every part shows correctly.

What to deliver

Three items, for a 6-second roadmap shot in the video and one PPT slide.

Deliverable	Format	Used for
Schematic	PNG export, 1920 × 1080	Video (2 s) + PPT
PCB layout, top view	PNG export, 1920 × 1080	Video (2 s) + PPT
3D render, rotating	Screen recording, 8–10 s	Video (2 s)

Caption for the video and slide: **"Phase 2: integrated DHVANI shield, replaces the breadboard (designed, not yet fabricated)."**